Stats 198 Project

Due Wednesday, April 22

Kina Desai, Allister Wolff, Florine Lijesen

Use of AI tools

Use

Introduction

Research question: Do higher income countries (measured by GDP per capita) have greater access to clean cooking fuels? - potential to go further:

Methodology

```r
library(tidyverse)
```

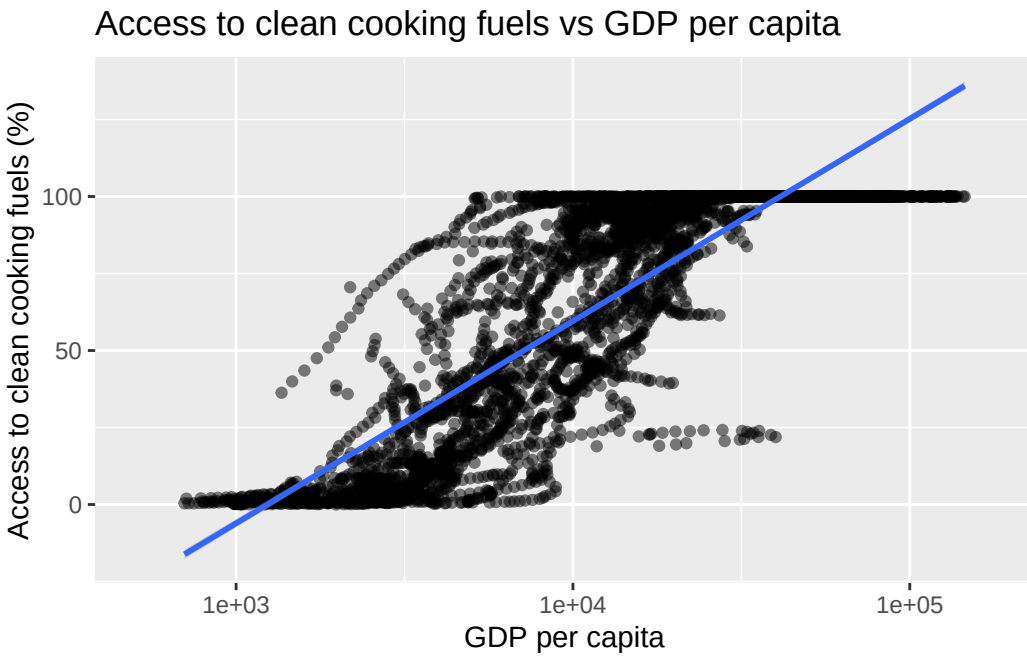
```
-- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
v dplyr     1.2.0     v readr     2.1.6
v forcats   1.0.1     v stringr   1.6.0
v ggplot2   4.0.2     v tibble    3.3.1
v lubridate 1.9.5     v tidyr     1.3.2
v purrr     1.2.1
-- Conflicts ------------------------------------------ tidyverse_conflicts() --
x dplyr::filter() masks stats::filter()
x dplyr::lag()    masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to becom
```

```r
data <- read_csv("access-to-clean-fuels-for-cooking-vs-gdp-per-capita.csv")
```

```
Rows: 8010 Columns: 7
-- Column specification --------------------------------------------------------
Delimiter: ","
chr (3): Entity, Code, World region according to OWID
dbl (4): Year, Access to clean fuels and technologies for cooking, GDP per c...

i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```r
ggplot(data, aes(
  x = `GDP per capita`,
  y = `Access to clean fuels and technologies for cooking`
)) +
    geom_point(alpha = 0.5) +
  geom_smooth(method = "lm") +
  scale_x_log10() +
  labs(
    x = "GDP per capita",
    y = "Access to clean cooking fuels (%)",
    title = "Access to clean cooking fuels vs GDP per capita"
  )
```

```
`geom_smooth()` using formula = 'y ~ x'

Warning: Removed 3632 rows containing non-finite outside the scale range
(`stat_smooth()`).

Warning: Removed 3632 rows containing missing values or values outside the scale range
(`geom_point()`).
```




```r
# Add income group column
data <- data |>
  mutate(income_group = case_when(
    `GDP per capita` < 1000  ~ "Low income",
    `GDP per capita` < 12000 ~ "Middle income",
    TRUE                     ~ "High income"
  ))|>
  filter(`Access to clean fuels and technologies for cooking` <= 100)  # remove bad values
```

Results

Discussion